

Loammi Code Tutorial - Python and R

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This document will help to explain what each script does, how to use it, and what the results mean. There are comments throughout the notebooks as well to help guide you through the analysis. Sample .csv data from the study is included, as well as relevant file outputs, but you can also use your own data as long as it matches the file formatting. Due to their large sizes, GeoTIFF rasters and shapefiles are NOT included in this GitHub; however, you may download them at the following links. Please note that you will need to convert the download files to GeoTIFF or shapefile formats using GIS software in order to use them with the provided scripts.

Florida Cooperative Land Cover Layer:

<https://myfwc.com/research/gis/wildlife/cooperative-land-cover/>

Click "Zipped Raster (~400MB)" to download (v 4.0, November 2025).

FNAI Conservation Lands:

<https://www.fnai.org/publications/gis-data>

Click "Feature Class in ZIP - Updated June 2026" to download.

For any questions, error troubleshooting or concerns, please reach out to avacjavacj@gmail.com.

1. LoammiOccs.R

Input files:

- All_Butterfly_iNat_Points.csv
- Loammi_Private_Coords_June.csv
- Skipper DNA samples.csv

This is the script used to generate our cleaned, filtered *A. loammi* occurrence data layer, the random background points layer, and the bias-weighted background points layer for the habitat index calculation. For the purposes of this paper, only the index based on the bias-weighted background layer was reported, and the random background layer was a point of comparison.

This script creates two cleaned, filtered occurrence data files: 1) one using a spatial thin of 150 m, which was used in the paper, and 2) one using a spatial thin of 1 km for comparison. This script also generates a plot showing the *A. loammi* occurrence points thinned by 150 m, the bias-weighted background points, and the accessible area estimated for *A. loammi* (Figure S1).

To run this script, you will need to set your working directory to correctly load the input files.

2. CLCRank.ipynb

Input files:

- Loammi_Thin_150m_June.csv (obtained from previous script)
- CLCv4State.tif (obtained from myFWC.com download link above)

To run using the occurrence data thinned by 1 km:

- Loammi_Thin_1km_June.csv (obtained from previous script)

To run this code, you will need:

- A .csv file of your occurrence points with latitude and longitude columns. The sample code has column names 'decimalLatitude' and 'decimalLongitude'. You can either set up your .csv to have these names and run the code as-is or modify the code where the comment specifies to update column names to whatever you choose.
- A GeoTIFF file in which the pixel values represent land cover types (e.g., a numeric value such as 1300). You may need to export this from a GIS software. Make sure the file path in the input configuration links directly to the .tif file and not just the folder it's in!

By default, this code will:

1. Reproject GPS points to match the projection of the raster.
2. Generate a 75-m buffer around each point to extract raster values. The buffer size can be changed if needed.
3. Calculate statistics and rankings based on both raw occurrence data and based on an index standardized by total area of each land cover type. It will output 4 .csv files:
 - landcover_counts, a file of each land cover type found within *all* the buffers and their respective proportions of the total buffer area;
 - landcover_proportions, a file that lists each *individual* buffer, its lat/long coordinates, the land cover types found within it and their respective area proportions within that buffer (this was not used in the paper but provides data for optional analysis);
 - landcover_rankings, a file that lists each land cover type, its proportion within the total buffer area (column name 'proportion', which is the same as the landcover_counts file), its proportion of the total raster ('overall_proportion'), its ranking based on occurrences ('RankOcc', calculated as its proportion of the total area of all buffers), its ranking based on an index calculation ('RankIndex', calculated by dividing the land cover proportions within the total buffer area by their proportions of the total raster), the number of buffers in which the land cover type covered the most area ('buffers_as_dominant'), the number of buffers in which the land cover type appeared ('buffer_count'), and finally, the raw pixel count of each land cover type within the total buffer area and total raster area ('buffer_pixels' and 'raster_pixels', respectively).

- NOTE: The “RankIndex” column generated by this script is NOT the one used in the paper; it reflects an older analysis that has since been replaced. The new index is calculated using a separate R script, as detailed later in this tutorial. You will not need to run the final cell to generate the landcover_rankings file if you are using the LoammiIndex.R script.
- Loammi_Points_With_Landcover: This file was created by extracting land cover types via point-based locations at individual pixels, rather than by proportions of buffers. This was not used in the study but may be helpful for other applications.

Each cell should run back-to-back. In the main configuration cell, you will need to update the paths to the input files and set up an output folder where the results will be stored.

3. LoammiIndex.R

Input files:

- landcover_counts_loammi.csv
(obtained by running CLCRank.ipynb with Loammi_Thin_150m_June.csv)
- landcover_counts_random.csv
(obtained by running CLCRank.ipynb with Loammi_Background_Random.csv)
- landcover_counts_weighted.csv
(obtained by running with Loammi_Bias_Weighted_Background_Points.csv)

To run using the occurrence data thinned by 1 km:

- landcover_counts_loammi_1km.csv
(obtained by running CLCRank.ipynb with Loammi_Thin_1km_June.csv)

Note: Input file names created manually after running previous script with alternate data sources.

To run this script, you will need to run the CLCRank code three times with each of the following GPS point layers: your main occurrence dataset (i.e., “Loammi_Thin_150m_June.csv”), a random background layer, and a bias-weighted background layer. This will generate the “landcover_counts” output file for each dataset. You will need to manually rename the output file each time to avoid overwriting the previously created output file. You will also need to set your working directory and adjust the file names as needed to correctly load the input files.

By default, this code will calculate two indices: one using the bias-weighted background layer and one using the random background layer. For this paper, only the weighted index was used. The new index was calculated with the following formula:

$$Habitat\ Index_x = \frac{(P_{L,x}/P_L)}{(P_{B,x}/P_B)}$$

$P_{L,x}$ = pixels of land cover type x within all Loammi buffers

P_L = total pixels within all Loammi buffers

$P_{B,x}$ = pixels of land cover type x within all background buffers

P_B = total pixels within all background buffers

This script also creates the file Top10Landcover.csv which contains a list of the top 10 land cover types ranked by the bias-weighted index, as well as lists of the land cover types within the top 10 that were classified as ‘Human’ or ‘Natural’ based on Kawula et al. (2025).

4. TopMaps.ipynb

Input files:

- CLCv4State.tif (obtained from myFWC.com download link above)
- Top10Landcover.csv (obtained from the previous script)
- Files from the FLMA shapefile download: flma.cpg, flma.dbf, flma.prj, flma.sbn, flma.sbx, flma.shp, flma.shx (obtained from FNAI.org download link above)

To run based on the occurrence data thinned by 1 km:

- Top10Landcover_1km.csv
(obtained by running LoammiIndex.R with landcover_counts_loammi_1km.csv)

Note: Top10Landcover_1km.csv file name created manually after running LoammiIndex.R with landcover_counts_loammi_1km.csv.

To run this code, you will need to set up an initial input folder that has the following files:

- A GeoTIFF land cover raster with pixel values representing land cover types,
- A shapefile of protected lands (or whatever area you wish to extract),
- A .csv file of land cover types you wish to map.

You will also need to set up an output folder where the resulting maps and data will be stored.

By default, this code will:

1. Extract the total areas of the land cover types listed in your .csv file and export as a GeoTIFF file (‘Top_Total.tif’).
2. Extract again the protected areas of those same land cover types using the conservation lands shapefile and export as another GeoTIFF file (‘Top_Protected.tif’).
3. Calculate the proportions protected of each land cover type and save as a .csv file (‘Top_Results.csv’).

The code is designed to avoid memory issues by processing the raster in chunks, since the raster used in the study was very large (statewide Florida). Larger rasters will take a long time to process, but the code can safely be left to run in the background as you do other things.

After the initial land cover extraction, you will need to update the path in the next cell to the newly made Top_Total.tif raster so that the code only extracts the protection portions of your selected land cover types and produces a raster titled Top_Protected.tif.

Finally, you will do the same in the last cell to calculate the protected portions of each land cover type. Simply update the paths with your new Protected and Total rasters where the comments prompt you to set raster paths.

5. DistanceAnalysisBuffer.ipynb

Input files:

- Loammi_Thin_150m_June.csv (obtained from previous script)
- Top10Landcover.csv (obtained from previous script)
- CLCv4State.tif (obtained from myFWC.com download link above)

This code assumes that you have some human land use (or other unusual land cover types) appearing within your top rankings, and you want to see how far away the specimens observed there were from another land cover type (such as a natural land cover type).

To run this code, you will need:

- A .csv file of occurrence points in lat and long columns,
- A .csv file of land cover codes sorted by 'Natural' or 'Human',
- A GeoTIFF raster file with those same land cover codes as pixel values.

By default, this code will:

1. Find and print input files (just as a sanity check).
2. Reproject GPS coordinates to match those of the GeoTIFF file and generate a 75-m buffer around each point to determine the dominant land cover type.
3. For each point, identify the land cover type containing the point (Point_Landcover) and calculate the Euclidean distance (Distance_meters) from the center of the raster cell containing the 75-m buffer center point to the center of the nearest raster cell representing one of the top 'Natural' land cover types (Nearest_Natural) listed in the Top10Landcover.csv file. To reduce memory requirements, the search function used progressively larger square search windows to identify the nearest 'Natural' land cover type. Because the search terminated when a 'Natural' land cover type was first encountered within a search window, the selected raster cell was not guaranteed to be the true nearest cell in all cases. Consequently, reported mean and maximum distances are conservative (i.e., true distances may be smaller). NOTE: When the nearest 'Natural' land cover type fell within the 75-m buffer, distances calculated were ≤ 75 m. The following script (LoammiFigures.R) subsets these data so only occurrence points with distances > 75 m to the nearest top 'Natural' land cover type are considered to fall outside a top 'Natural' land cover type (75 m corresponds with GPS uncertainty in our dataset).

4. Generate two .csv files, “all_points” and “human_land_use_points” (same analysis but lists only the points that fell within the ‘Human’ land cover types designated in the Top10Landcover.csv file). NOTE: The latter .csv file was not used for the paper.

6. LoammiFigures.R

Input files: (all obtained from the previous scripts)

- Index.csv
- Top_Results.csv
- CLCStateTable.csv
- all_points.csv
- Index_1km.csv
- Top_Results_1km.csv

This is the final script that visualizes your data!

To run this code, you will need:

- A .csv file of your land cover rankings (i.e., “Index.csv”),
- A .csv file of the protected proportions of your top habitats (i.e., “Top_Results.csv”),
- A .csv file of land cover types matching codes to names (i.e., “CLCStateTable.csv”),
- A .csv file of your buffer distances (i.e., “all_points.csv”),
- A .csv file of your land cover rankings based on the 1-km thin (i.e., “Index_1km.csv”),
- A .csv file of the protected proportions of your top habitats based on the 1-km thin (i.e., “Top_Results_1km.csv”)

If you have a .csv file that matches the pixel land cover codes with their names, this script will automatically convert the numeric codes in the datasets to names before plotting. Otherwise, you can modify the code to use the ‘landcover_type’ pixel code values.

By default, this code will generate six plots:

1. Bar chart showing the top land cover types based on the bias-weighted index,
2. Bar chart of those same land cover types showing their protected proportions,
3. A combined plot showing both charts side by side,
4. A table that displays results for the 150-m spatial thin,
5. A table that displays results for the 1-km spatial thin,
6. A table that displays the occurrence points classified as outside the top ‘Natural’ habitats.